e) Data flows between a database and its data files.

5. Given the following scenario: A rogue system administrator can turn off audit logging before performing actions using an API. Which of the STRIDE threats being abused in this case? Recall from section 1.1 that an audit log records who did what on the system.

1.5 Security mechanisms

Threats can be countered by applying security mechanisms that ensure that particular security goals are met. In this section we will run through the most common security mechanisms that you will generally find in every well-designed API:

- *Authentication* is the process of ensuring that your users and clients are who they say they are.
- *Access control* (also known as *authorization*) is the process of ensuring that every request made to your API is appropriately authorized.
- *Audit logging* is used to ensure that all operations are recorded to allow accountability and proper monitoring of the API.
- *Rate-limiting* is used to prevent any one user (or group of users) using all of the resources and preventing access for legitimate users.

Figure 1.7 shows how these four processes are typically layered as a series of filters that a request passes through before it is processed by the core logic of your API. As